

Himadri Mishra

AI Systems and Developer Infrastructure Engineer

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SUMMARY

Senior AI engineer (IIT-BHU, 8 years) building agent-facing tools, production LLM workflows and ML platforms. I help teams ship reliable AI product systems with explicit orchestration, evaluation, recovery and handover-ready artifacts.

EXPERIENCE

Current engineering work

- Aligned CLI/MCP/SDK interfaces: tool descriptions, documentation navigation, routing and API guidance for developer and agent workflows.
- Built SDK client methods and regression tests to keep behavior consistent across client surfaces.
- Added browser-session entry points and repaired runtime request/recovery behavior, with application and proxy tests.
- Migrated an MCP server runtime and hardened CLI credential setup, with smoke and regression tests.
- Implemented retrieval-backed draft generation, claim checks and human-review gates, with regression tests.

Employment history

Senior AI Engineer

Knit

Remote, India

May 2025 – April 2026

- Architected **Knit 3.0 research reporting platform**: survey ingestion to verified insights, charts, memos and PPTX decks, **reducing post-fielding turnaround from 48–72 hours to under 1 hour**.
- Built a **sandboxed insight engine**: LLM-generated Python analyses across 30–50 tasks per report, each verified by independent judge agents.
- Designed **DAG-based orchestration** with dependency-aware parallel execution and a visualization pipeline producing 15–25 Highcharts charts per report with multi-threshold quality scoring.
- Unified reporting, questionnaire, ingestion and deck agents on a **shared Python platform**: multi-provider LLM routing, pgvector RAG, OpenTelemetry, Langfuse, SSE and generated REST APIs.
- Drove system architecture as a senior IC for India AI workflows, coordinating design decisions across research reporting, questionnaire, ingestion and deck-generation agents.

Independent ML Engineer (Freelance)

Stealth Product Team

Remote, India

March 2025 – April 2025

- Built an **end-to-end text-to-vector-icon generation pipeline**, covering dataset creation, model benchmarking, LoRA fine-tuning, segmentation and vectorization.
- Curated **18k vector icons** and generated high-quality descriptions through batch Gemini API calls; benchmarked proprietary and open-source text-to-image models including Flux and SDXL.
- Trained and evaluated thin-structure segmentation models including U-Net, U-Net++, U2-Net and SegFormer, then built a graph-based skeleton-to-vector conversion workflow.

Career Break

Personal sabbatical

May 2024 – February 2025

Senior Research Engineer
Epic! for Kids

Remote, India
February 2023 – May 2024

- Owned ML pipeline and infrastructure across discovery, recommendation and search; led Elasticsearch autocomplete revamp, outperforming the prior solution in 80%+ cases.
- Cut Docker build time by 50%, Kubernetes pod usage by 100x, spot instance errors by 99% and platform cost by 10x; shipped ML features and cross-stack analytics contributions.

Senior Research Engineer
Tangible Play (Osmo)

Remote, India
September 2019 – February 2023

- Led computer vision across India and US teams for Worksheets; raised accuracy from 93% to 98% and built a real-time U-Net shaded-region model achieving 80% IoU.
- Led tracing-the-dots games, boosting engagement by 20%; prototyped 10fps teacher-feedback CV and automated tagging workflows, cutting manual effort by 99%.

Deep Learning Engineer
Whodat (AR startup, acquired by Byju's)

Bangalore, India
July 2018 – August 2019

- Built a fast ORB detector in C++, **20% faster than ORB-SLAM**; researched monocular depth estimation with DeMoN and MvDepthNet. *Team transitioned to Tangible Play/Osmo post-acquisition.*

SKILLS

AI systems: Agentic workflows, MCP, CLI tools, SDK/API design, LLM evaluation, RAG, guardrails, source-grounded assistants

ML and data: PyTorch, TensorFlow, computer vision, NLP, survey analytics, MLOps, statistics, sentence transformers

Platform: Python, TypeScript, C++, Java, Go, Dart, PHP, FastAPI, REST, SSE, PostgreSQL, pgvector, Redis, S3, Elasticsearch, PySpark, Docker, Kubernetes, CI/CD, Google Cloud, OpenTelemetry, Langfuse

Web: Next.js, React, Django, Angular

EDUCATION AND AWARDS

Indian Institute of Technology (IIT-BHU), Varanasi 2013 – 2018
Integrated Dual Degree in Computer Science and Engineering GPA: **9.28/10**

Kaggle FIDE and Google Efficient Chess AI Challenge (2025): top 6% globally, Rank 60 of 1120.

Research Intern, UC Berkeley (2017): neural programmer-interpreters under Professor Dawn Song, SN Bose Scholar Program Awardee.

Software Engineering Intern, Microsoft (2016): dialog-based chatbots for the Incentive Compensation team.